FERNS N PETALS

PROJECT SUMMARY

Four festival months carry 68% of the year's revenue.

A sales analysis of 1,000 orders placed with an Indian gifting retailer across 2023, delivered as an interactive Excel dashboard and a self-contained web dashboard.

PERIOD	GRAIN	SOURCES	RECORDS
1 Jan – 29 Dec 2023	One row per order	3 CSV files	1,000 / 100 / 70

TOTAL REVENUE
₹35,20,984
3,045 units across 1,000 orders

AVERAGE ORDER VALUE
₹3,521
3.05 units per basket

REVENUE PER CUSTOMER
₹35,210
100 active buyers

AVERAGE DELIVERY
5.5 days
64% arrive within a week

PEAK MONTH
August
₹7,37,389 · Raksha Bandhan

REACH
301 cities
Delivery, not billing, addresses

OBJECTIVE

What the analysis set out to answer

Three raw CSV extracts — customers, orders and products — with no revenue column and no stated relationships. The task was to turn them into something a commercial team could act on: where revenue actually comes from, which occasions and products earn rather than merely sell, and whether the delivery promise holds.

The answer reframed the business. Demand here is not a smooth curve with seasonal bumps; it is four narrow windows carrying two-thirds of the year. That makes inventory, staffing and courier capacity a **calendar problem before it is a demand problem** — the finding every other recommendation depends on.

Scope

Orders	1,000
Customers	100
Catalogue items	70
Categories	7
Occasions	7
Delivery cities	301

Deliverables

- Interactive Excel workbook, 8 sheets
- Self-contained web dashboard, 17 views
- Field-level data dictionary
- Python script reproducing every figure

METHOD

How the data was built

01

Extraction

Three CSV files imported without modification and kept as the immutable base layer, so every downstream figure traces back to a source row.

02

Cleaning

Dates parsed explicitly as `DD-MM-YYYY` rather than left to the default parser, which silently reads 07-11-2023 as 11 July and shifts the entire monthly trend. Multi-line address fields normalised; time columns typed so they combine with dates for hour-level analysis.

03

Modelling

A flat fact table joining orders to products on `Product_ID` and to customers on `Customer_ID`, with fifteen derived columns — revenue, lead time, month, hour, weekday, price band, delivery bucket — each a live formula rather than a pasted value.

04

Analysis

Aggregations built with `SUMIFS`, `COUNTIFS` and `AVERAGEIFS` across occasion, category, month, hour, weekday, price band, basket size, city and customer. Rankings resolved with `LARGE` plus a tie-break key so ties settle by revenue rather than arbitrarily.

05

Dashboard

Two dropdowns drive eleven charts, six KPI tiles, both ranking tables and the category matrix. Selecting `(All)` resolves to the wildcard `"*"`, so a single formula pattern serves the filtered and unfiltered case alike.

MODEL

Two relationships, one derived measure

`Revenue = orders.Quantity × products.[Price (INR)] — no source file contains a revenue column.`

DESIGN NOTE

Why formulas, not a PivotCache

The workbook deliberately avoids PivotTables and a Power Pivot data model, reproducing slicer behaviour with data validation and wildcard criteria instead. That keeps three properties which matter for a shared analytical file: every calculation is visible in the cell that produces it; no hidden cache can fall out of step with its source; and it recalculates identically in Excel, LibreOffice and Google Sheets.

> The trade-off is honest: at this grain — 1,000 rows, roughly 17,000 formulas — recalculation is instant. At a million rows the Power Pivot engine would be the right call, and the same model would port to it directly.

FINDINGS

What the data actually says

4.3×

FESTIVAL UPLIFT

The year is four windows, not twelve months

February, March, August and November — Valentine's, Holi, Raksha Bandhan and Diwali — average ₹6,00,722 against ₹1,39,762 in a quiet month, and carry 68% of annual revenue between them.

35.9%

FROM 5-UNIT BASKETS

Bundle size is the strongest lever on revenue

Five-unit orders are 21.9% of volume but 35.9% of revenue. Single-unit orders are a near-identical 19.6% of volume and just 6.4% of revenue — so basket size, not traffic, is where order value moves.

₹4,785

TOP OCCASION AOV

Volume and value point at different occasions

Anniversary brings the most orders at 205, but Raksha Bandhan earns the highest average order value — 75% above Birthday, the weakest. Ranking occasions by order count alone inverts the priority.

80%

REVENUE ABOVE ₹1,000

The premium half of the catalogue does the earning

Items priced over ₹1,000 account for 60% of orders but 80% of revenue. Colors is the largest category at ₹10,05,645; Soft Toys and Sweets follow at roughly ₹7.4 lakh each.

0.1%

SHIP TO BUYER'S CITY

Gifts do not travel to the buyer

The delivery city matches the buyer's own city in one order in a thousand — 301 delivery cities against 86 billing cities. Regional analysis built on the customer address would describe the wrong map entirely.

46

ORDERS PAST 10 DAYS

The delivery tail matters more than the mean

Delivery averages 5.5 days and 64% land inside a week, but 46 orders take more than ten. For a dated occasion a late gift has failed outright, so the tail — not the average — is the service risk.

DETAIL

Revenue by occasion

OCCASION	REVENUE	ORDERS	AVG ORDER VALUE
Anniversary	₹6,74,634	205	₹3,291
Raksha Bandhan	₹6,31,585	132	₹4,785
All Occasions	₹5,86,176	126	₹4,652
Holi	₹5,74,682	180	₹3,193
Birthday	₹4,08,194	149	₹2,740
Valentine's Day	₹3,31,930	113	₹2,937
Diwali	₹3,13,783	95	₹3,303

DATA QUALITY

Traps found in the source files

Each produces a plausible-looking but wrong answer if missed, which is what makes them worth documenting rather than quietly fixing.

ISSUE	CONSEQUENCE IF MISSED	RESOLUTION
Product names not unique	Quia Gift is IDs 21 and 56, Error Gift is 65 and 70 — different categories and prices. Grouping by name inflates one row to ₹1,14,476 and pushes two real top-ten products off the chart.	`Aggregate on Product_ID`
Dates are DD-MM-YYYY	Default parsing reads 07-11-2023 as 11 July, shifting the monthly trend that the headline finding rests on.	`Parse format explicitly`
Two Occasion columns	Orders carry the occasion the gift was bought for; products carry a merchandising tag. They disagree often.	`Use the order column only`
Delivery vs billing city	Location is the recipient's city, not the buyer's. Regional reporting on the wrong column describes a different market.	`Documented, both retained`
Deliveries cross the year	Six late-December orders deliver in January 2024 and vanish if the delivery date is filtered to 2023.	`Filter on order date`
Contact_Number stored as integer	Any leading zero is already lost in the source file and cannot be recovered.	`Flagged, treat as text`

No missing values and no duplicate order keys across the three files.

OUTPUT

What was delivered

Excel workbook

Eight sheets, eleven charts, two dropdown filters, 16,979 formulas and no hardcoded results — edit a price and the whole dashboard follows.

`Dashboard` `Data Dictionary` `Calcs` `Model` `3 source tabs` `Lists`

Web dashboard

One HTML file, no build step and no dependencies. Seventeen views with live filtering, hover detail and a light/dark theme.

`Hand-written SVG` `Embedded data` `Client-side` `Responsive`

VERIFICATION

Every figure is reproducible

A Python script recomputes every number in this document from the three source CSVs, and its output reconciles exactly with the workbook. The analysis can be audited in one command rather than taken on trust.

> The dataset is a sample used for analysis practice: product names and descriptions are placeholder text and customer records are generated. The structural findings — festival seasonality, basket-size economics, the delivery-city gap — hold for the data as given; product-level results should be read as method rather than a view of the real catalogue.

TOOLS

Built with

Excel — data model, formula layer, charting, data validation, conditional formatting. **Python (pandas)** — independent verification of every published figure. **HTML, SVG, vanilla JavaScript** — the web dashboard, written without a charting library.